

Strips used in Glucometer a chemical called glucose oxidase which reacts with the glucose in the blood sample and is converts it into an acid called gluconic acid.

4.5 Tricarboxylic Acid Cycle

TCA cycle was first elucidated by in 1937 "Sir Hans Adolf Krebs, a Germen biochemist. It is also known as Tricarboxylic acid cycle, Citric acid cycle or Amphibolic cycle. In prokaryotic cells, the citric acid cycle occurs in the cytoplasm; in eukaryotic cells it takes place in the matrix of the mitochondria.

The process oxidizes glucose derivatives, fatty acids, and amino acids to carbon dioxide (CO_2) through a series of enzyme controlled steps. The purpose of the Krebs cycle is to collect high energy electrons from these fuels by oxidizing them, which are transported by activated electron carriers such as NADH and $FADH_2$ to electron transport chain. The Krebs cycle is also the source for the precursor of many other molecules and is therefore an amphibolic pathway (both anabolic and catabolic reactions take place in this cycle) and therefore, it can be used for both the synthesis and degradation of bio molecules.

$$\text{Pyruvate} + \text{CoA-SH} + \text{NAD}^+ \xrightarrow{\text{Pyruvate dehydrogenase complex}} \text{Acetyl CoA} + \text{Co}_2 + \text{NADH}$$

Pyruvate cannot enter the Krebs cycle directly. In a preparatory step, it must loses one molecule of CO_2 and becomes a two-carbon compound. This process is called decarboxylation. The two-carbon compound, called acetyl group, attaches to coenzyme A through a high-energy bond, the resulting is a complex known as acetyl coenzyme (acetyl CoA). During this reaction, pyruvic acid is also oxidized and NAD^+ is reduced to NADH by pyruvate dehydrogenase complex. This multi enzyme complex is responsible for the conversion of pyruvate to acetyl-coA. Therefore (PDHC) contribute to linking the Glycolysis metabolic pathway to the citric acid pathway.



Pyruvate dehydrogenase deficiency is a common cause of lactic acidosis in new born and often present with poor feeding.

Pyruvate dehydrogenase complex is a complex of three enzymes that convert pyruvate into acetyl-coA by a process called pyruvate decarboxylation (Figure 4.5). Acetyl-CoA may then be used in the citric acid cycle to carryout cellular respiration, and this complex links the Glycolysis metabolic pathway to the TCA cycle.

The Krebs cycle generates a pool of chemical energy (ATP, NADH, and $FADH_2$) from the oxidation of Pyruvic acid and it loses one carbon atom as CO_2 and reduces NAD^+ to NADH. The resulting two carbon acetyl molecule is joined to Co enzyme A. Acetyl CoA transfers its acetyl group to a 4C compound (oxaloactate) to make a 6C compound (Citrate) and the Coenzyme A is released which goes back to the link reaction to form another molecule of acetyl CoA. Oxaloacetate is both the first reactant and the product of the metabolic pathway (creating a loop).

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